

Random Vectors.

Say, we are interested in two (or more) r.v.s as a

PAIR (or VECTOR), i.e.,

we look @ (Y_1, Y_2) (or $(Y_1, Y_2, \dots, Y_n)$).

Then, it's not sufficient to only look @ their "individual" dist'n, but also their ASSOCIATION.

Example. $Y_i \dots$ cointoss for $i=1,2$ of fair coins

independence

$$\begin{array}{l} \{Y_1=H, Y_2=H\} \\ \{Y_1=H, Y_2=T\} \end{array} \quad \begin{array}{l} \{Y_1=T, Y_2=H\} \\ \{Y_1=T, Y_2=T\} \end{array}$$

complete dependence

$$\begin{array}{l} \{Y_1=H, Y_2=H\} \\ \{Y_1=T, Y_2=T\} \end{array}$$

A Modification Imputing Dependence

Experiment:

$$\left\{ \begin{array}{l} \text{IF } Y_1=H, \text{ then } Y_2 \sim \begin{cases} H & \text{prob. } 0.9 \\ T & \text{prob. } 0.1 \end{cases} \\ \text{IF } Y_1=T, \text{ then } Y_2 \sim \begin{cases} H & \text{prob. } 0.5 \\ T & \text{prob. } 0.5 \end{cases} \end{array} \right.$$

Discrete 2D Environment.

The Joint Distribution Table.

$X \backslash Y$	y_1	y_2	...	y_j	...	y_l
x_1						
x_2						
$\vdots$						
x_i				p_{ij}		
$\vdots$						
x_m						

The MARGINAL DISTRIBUTION of X

$$p_X(x_1) = \sum_{j=1}^l p_{1j}$$

$$\vdots$$

$$p_X(x_i) = \sum_{j=1}^l p_{ij}$$

The Marginal Dist'n of Y

$\downarrow$
 $\rightarrow 1$

$p_{ij} = P[X=x_i, Y=y_j]$, i.e., the joint pmf of (X, Y) for all x_i, y_j

X and Y are independent
iff

$$p_{ij} = p_X(x_i) \cdot p_Y(y_j) \text{ for all } i, j$$

Example. We independently throw two fair dice and record the results as (Y_1, Y_2)

joint pmf: $p_{ij} = \frac{1}{36}$ for all $1 \leq i, j \leq 6$

Define $W = Y_1 + Y_2$

Q: What is the joint dist'n table for (Y_1, W) ?

$Y_1 \backslash W$	2	3	4	5	6	7	8	9	10	11	12	
1							0	0	0	0	0	$\frac{1}{6}$
2	0							0	0	0	0	$\frac{1}{6}$
3	0	0							0	0	0	$\frac{1}{6}$
4	0	0	0							0	0	$\frac{1}{6}$
5	0	0	0	0							0	$\frac{1}{6}$
6	0	0	0	0	0							$\frac{1}{6}$
	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	

$$p_{44} = 0 \neq$$

$$p_{Y_1}(4) \cdot p_W(4) = \frac{1}{6} \cdot \frac{3}{36}$$

The Continuous Case.

Recall: For a continuous r.v. Y w/ a pdf f_Y , we have

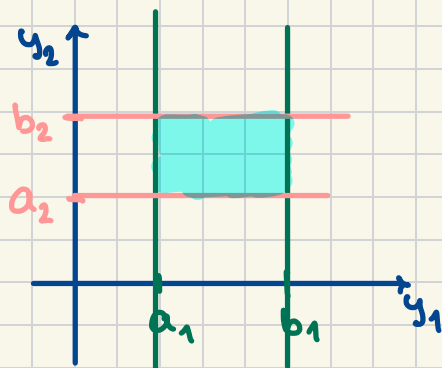
$$\begin{aligned}\mathbb{P}[Y \in [a, b]] &= \mathbb{P}[a \leq Y \leq b] \\ &= \int_a^b f_Y(y) dy \quad \text{for all } a \leq b\end{aligned}$$

In multiple dimensions

Consider a r.v. $(Y_1, \dots, Y_n)$ which is jointly continuous w/ joint density $f_{Y_1, \dots, Y_n} : \mathbb{R}^n \rightarrow [0, \infty)$

Then,

$$\begin{aligned}\mathbb{P}[Y_1 \in [a_1, b_1], Y_2 \in [a_2, b_2], \dots, Y_n \in [a_n, b_n]] &= \\ &= \int_{a_1}^{b_1} \int_{a_2}^{b_2} \dots \int_{a_n}^{b_n} f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) dy_n \dots dy_2 dy_1\end{aligned}$$



For any "nice" region $A \subseteq \mathbb{R}^n$:

$$P[(Y_1, \dots, Y_n) \in A] = \underbrace{\int \int \dots \int}_A f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) dy_n dy_{n-1} \dots dy_2 dy_1$$